

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 6 Summer 2025 (Regular)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTCVPE604A - INDUSTRIAL WASTE TREATMENT

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question)

12

- 1 A factory discharges 800 kg of BOD5 per day. What is the population equivalent (PE) considering per capita BOD5 = 80 g/day?
A. 800
B. 10,000
C. 1,000
D. 6,400
- One of the major factors influencing the flow pattern of industrial wastewater is: _____
A. Daily rainfall pattern
B. Seasonal temperature changes
C. Shift operation in factories
D. Availability of power supply
- 3 What is the molecular weight of benzene (C_6H_6) used for ThOD calculations?
A. 70
B. 80
C. 92
D. 78
- 4 The first step in minimizing the effects of industrial wastes is to reduce the: _____
A. Temperature
B. pH level
C. Volume
D. Odor
- 5 Equalization is used to: _____
A. Increase sludge generation
B. Increase effluent temperature
C. Minimize fluctuations in wastewater characteristics and flow
D. Eliminate need for treatment

- 6 Proportioning of industrial waste aims to: _____
 A. Reduce temperature of effluents
 B. Eliminate all effluent pollutants
 C. Match waste discharge with municipal sewage flow
 D. Increase alkalinity of wastewater
- 7 What is the primary function of the selective membrane in a reverse osmosis (RO) system?
 A. To dissolve salts in wastewater
 B. To allow only large molecules to pass
 C. To neutralize acidic waste
 D. To allow only water molecules to pass
- 8 Physical adsorption involves: _____
 A. Van der Waals forces and multilayer formation
 B. High specificity and irreversible layers
 C. Chemical bonding
 D. Use of acids and bases for reaction
- 9 The high COD in pulp and paper mill waste is mainly due to the presence of: _____
 A. Proteins
 B. Lignin
 C. Chlorine
 D. Glucose
- 10 Oil and grease content in dairy waste is primarily due to: _____
 A. Milk solids
 B. Butter and cream processing
 C. Milk pasteurization
 D. Ice cream mixing
- 11 Which of the following is a solid waste from the sugar industry used in boilers?
 A. Press mud
 B. Molasses
 C. Bagasse
 D. Clarified juice
- 12 If a water body has a WQI value of **96**, it is categorized as: _____
 A. Excellent
 B. Good
 C. Poor
 D. Very Poor

Q2. Solve the following.

- A) Define and differentiate between **Grab and Composite sampling** with examples. Also state when each type of sampling is preferred. 6
- B) Define "**Designated Best Use Classification**" of surface waters. Explain water quality criteria for Class A, B, C, D and F Class water as per CPCB. 6

Q3. Solve the following.

- A) List the methods of **waste volume reduction** in industries. Explain any one method in detail. Support your answer with examples. 6
- B) A Industry wants to implement **Zero Liquid Discharge (ZLD)**. As an civil engineering expert guide industry about ZLD process in detail 6

Q4. Solve Any Two of the following.

- A) Draw a simplified treatment flowsheet for a **textile industry** handling colored and alkaline wastewater. Indicate key treatment steps. 6
- B) Draw a neat manufacturing process flow sheet of a **Sugar Industry** and clearly indicate sources of wastewater generation. 6
- C) Provide a tabulated summary of wastewater characteristics of the **Pulp and Paper Industry** and suggest a treatment flow.sheet for combined waste. 6

Q5. Solve Any Two of the following.

- A) Define and explain the concept of **Common Effluent Treatment Plants (CETPs)**. What are their objectives, and why are they crucial for small and medium scale industries? 6
- B) Discuss constitution of state board according to '**The Water Act**' 6
- C) Discuss Socio-Economic implications of water pollution 6

Q6. Solve Any Two of the following.

- A) i. A factory discharges 400 kg/day of BOD. Calculate its **Population Equivalent (PE)** using a per capita BOD contribution of 80 g/day. Explain the significance of PE in designing treatment plants. 6
- ii. Calculate the **Treatability Index (TI)** for a wastewater sample with BOD = 150 mg/L and COD = 300 mg/L. Interpret the result and recommend a suitable treatment method.

- B) A water sample was tested for four water quality parameters. The observed values are given below, along with their **IS 10500:2012 permissible limits** and **ideal values** 6
- Using the **Arithmetic Weighted Mean Method**, calculate the **Water Quality Index (WQI)** and classify the water quality.

Parameter	Observed Value	Standard Value	Ideal Value
pH	7.6	8.5	7.0
TDS (mg/L)	720	1000	0
Nitrate (mg/L)	30	45	0
Iron (mg/L)	0.5	0.3	0

- C) Explain the concept of **ISO 14000**. How does **ISO 14001** certification help industries in improving environmental performance and achieving sustainable development? 6

*** End ***